

Iris Dataset Exploratory Data Analysis Report

Objective

To perform Exploratory Data Analysis (EDA) on the Iris dataset using Python libraries to understand data patterns, relationships, and correlations.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Google Colab

Dataset

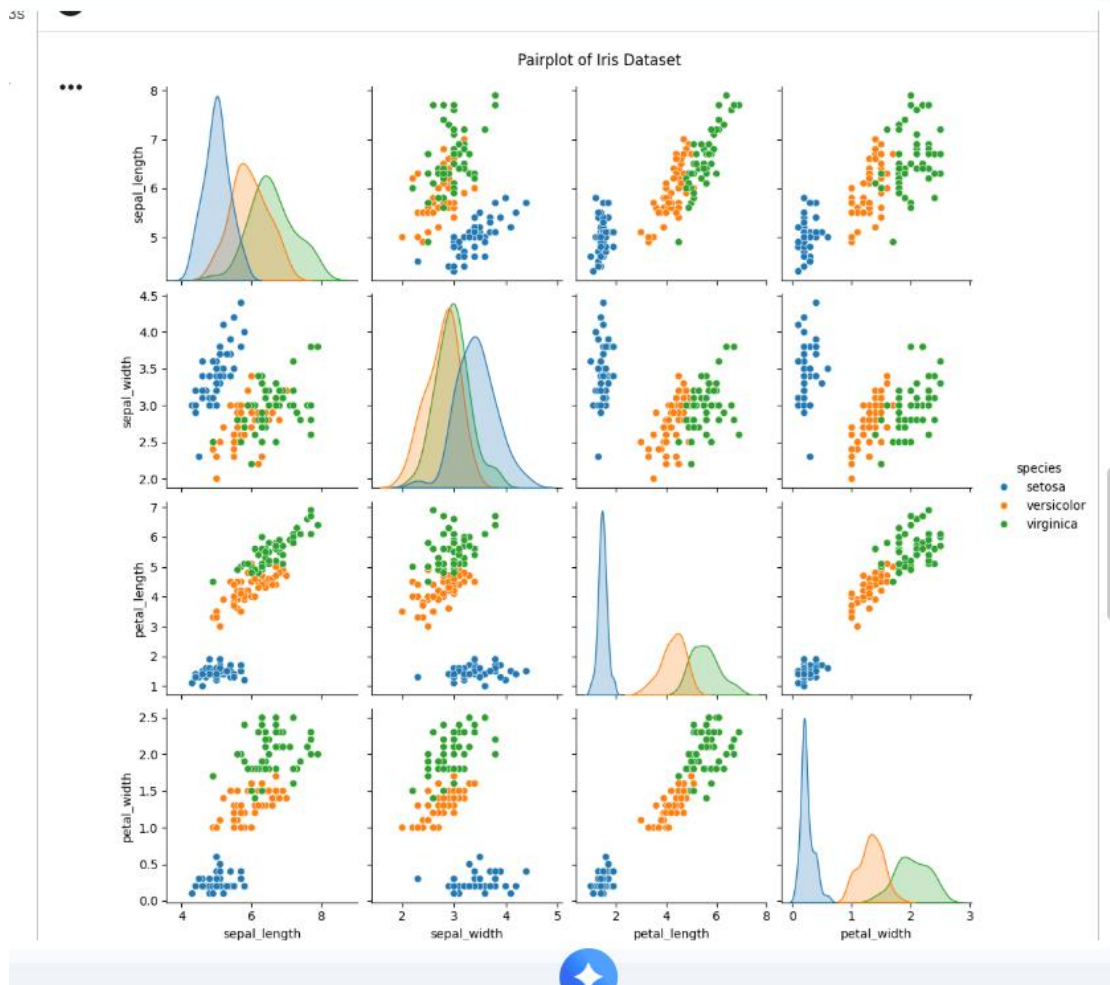
The Iris dataset contains flower measurements from three species:

- Setosa
- Versicolor
- Virginica

Tasks Performed

- Loaded the dataset
- Generated summary statistics
- Checked missing values
- Created pairplot visualization
- Generated correlation heatmap
- Identified feature relationships and patterns

Pairplot Analysis:

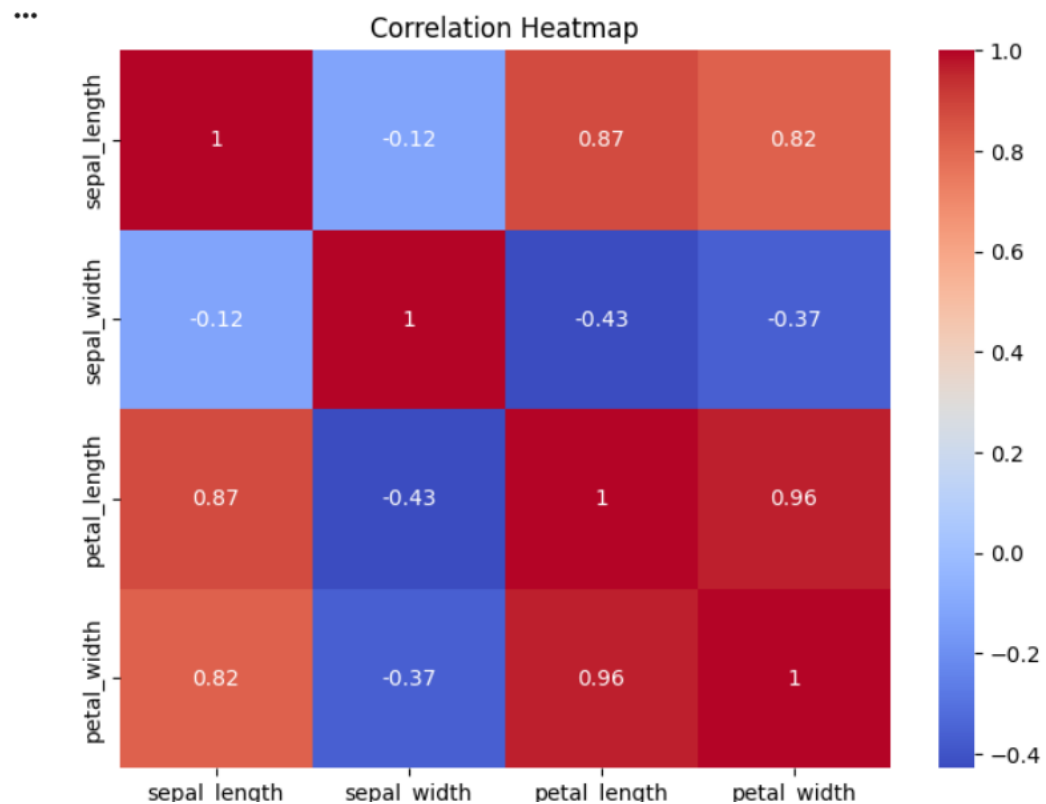


Observations:

- Setosa species is clearly separated from other species.
- Versicolor and Virginica show slight overlap.
- Petal length and petal width show strong relationships.

Correlation Heatmap Analysis:

```
map(df.select_dtypes(include='number').corr(),annot=True,  
:("Correlation Heatmap")  
:)
```



Observations:

- Petal length and petal width have strong positive correlation.
- Sepal length is positively correlated with petal length.
- Sepal width shows weaker correlation with other features.

Final Outcome

Successfully analyzed the Iris dataset and identified important patterns, feature relationships, and correlations using visualizations.